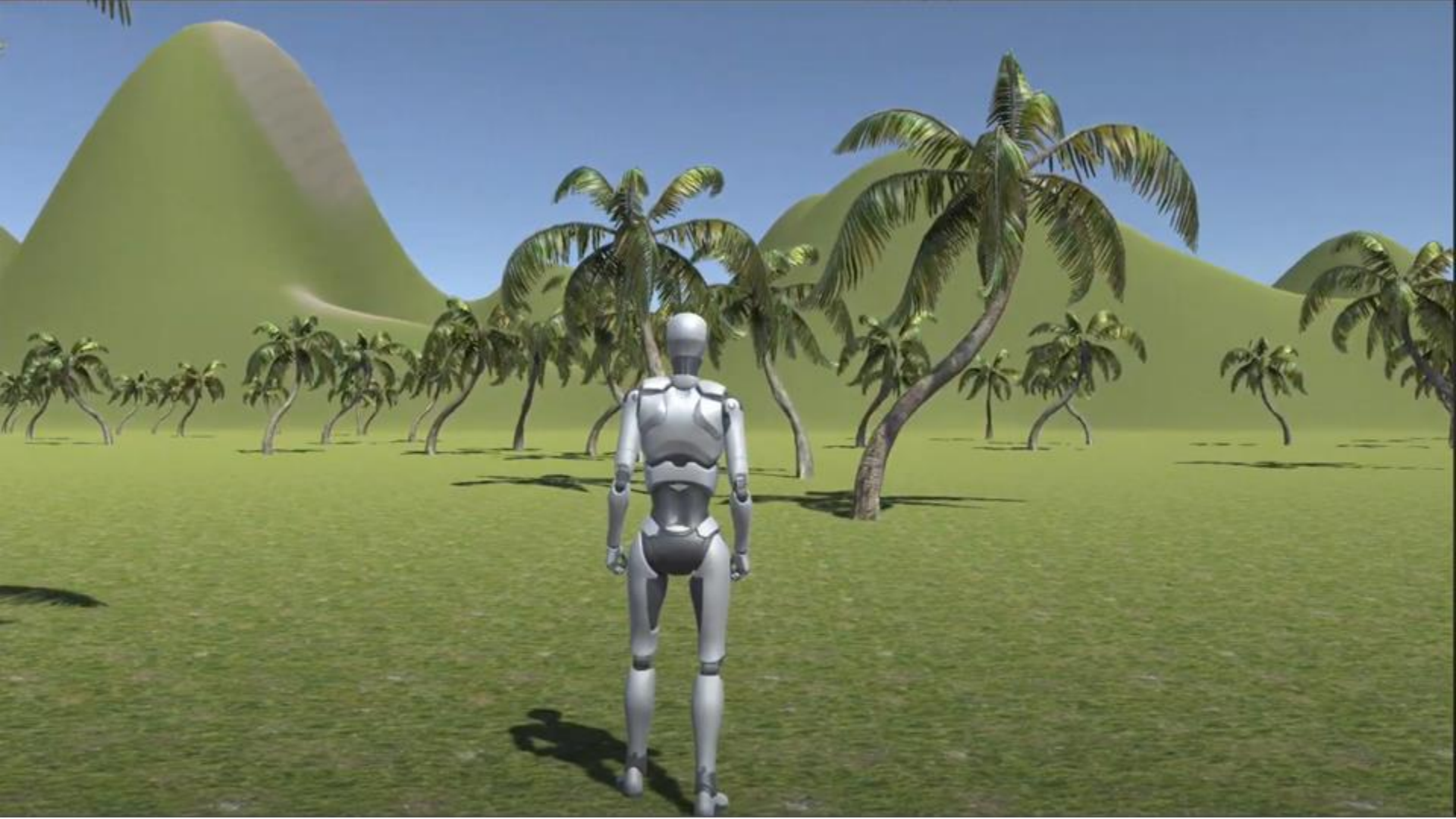


Complex System

Week 02 - Intelligent Agents



Agent

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Definition: Entities/systems that **perceive** their environment, **make decisions**, and **take actions** to **achieve specific goals** or objectives.

Types:

Physical Agents: Robots, autonomous vehicles, drones.

Software Agents: Programs running on computers or virtual entities.



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Characteristics

Sensing: Equipped with sensors to gather information

Processing: Analyze and process information

Decision-Making: Make decisions based on analysis and predefined rules

Action: Initiate actions or responses based on decisions.

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Applications: Used in AI and multi-agent systems, Model and simulate behaviors in complex environments, Interactions among agents lead to emergent behaviors and patterns, Eg: **Virtual Game Character**



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Virtual game character is an agent. "Sees" its surroundings in the game world (e.g., detects treasure chests). Follows predefined rules (e.g., moves toward a detected treasure chest). Collects the treasure and adds it to its inventory.



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Agents, whether physical or software-based, are autonomous entities that perceive, decide, and act in various applications, from gaming to complex AI systems. Their interactions can result in emergent behaviors, a fundamental concept in complex systems and AI.



BUT IS THIS AN “INTELLIGENT” AGENT?



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Definition: Intelligent agents possess advanced capabilities in AI and computer science beyond basic agents. They exhibit a higher degree of autonomy and adaptability. Eg: Autonomous car

Key Characteristics

- **Autonomy:** Operate independently, perceive the environment, process information, and respond without continuous human intervention.
- **Learning and Adaptation:** Learn from experiences, adapt to changing conditions, and improve performance over time.
- **Goal-Oriented:** Driven by goals or objectives, work toward specific tasks or optimization criteria.

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- **Sensing and Perception:** Perceive and interpret information from the environment using various sensors.

- **Reasoning and Decision-Making:** Make complex decisions based on available information and handle uncertainty.

- **Communication:** Communicate with other agents or systems, enabling collaboration.

- **Adaptability to Novel Situations:** Handle new situations by generalizing from learning and problem-solving capabilities.

Examples: Virtual personal assistants (e.g., Siri, Alexa), autonomous robots, self-driving cars, AI-powered chatbots.

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Practical Scenario: Interaction with a virtual personal assistant like "Alexa".

- **Autonomy:** Activates upon hearing the wake word.
- **Learning and Adaptation:** Recognizes your voice and improves understanding over time.
- **Goal-Oriented:** Aims to provide accurate weather or other information.
- **Sensing and Perception:** Processes spoken query using natural language understanding.

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- **Reasoning and Decision-Making:** Accesses weather and traffic data, determines location, and communicates the response.

- **Communication:** Responds with spoken weather information.

- **Adaptability to Novel Situations:** Handles follow-up questions and continues the conversation.

Intelligent agents, like virtual personal assistants, showcase advanced AI capabilities, enhancing user interactions and simplifying tasks through autonomous learning and adaptive decision-making.

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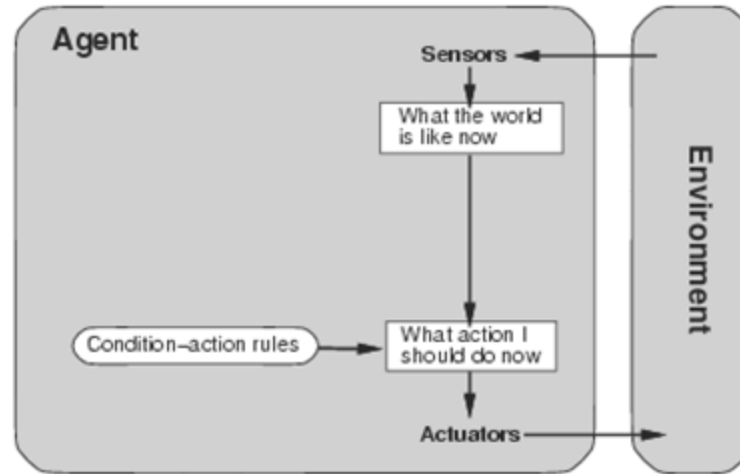


Fig: Schematic diagram of a simple reflex agent. (Ref: Artificial Intelligence: A Modern Approach, S Russell, P Norvig, page 47)

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- **Agent Rationality:** An AI agent aims to act rationally, making optimal choices based on circumstances.

- **Performance Measure:** To assess an agent's performance objectively, designers define a performance measure.

- **Defining Success:** The performance measure specifies success criteria for assigned tasks.

Example: "Cut the grass" task:

Performance Measure: "Has all the grass been cut?"

Rational Agent: Selects actions that maximize its performance measure for any percept sequence.

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Goals of an Agent:

- **Rationality:** Make informed choices given information.
- **Optimization:** Achieve the best possible outcomes.
- **High Performance:** Meet or exceed performance measure criteria.

In the design and evaluation of intelligent agents, rationality, optimization, and high performance are key goals to ensure effective decision-making and task execution in various domains, including AI and robotics.

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PEAS Description: Autonomous Taxi

Performance Measure (P): Safely and efficiently transport passengers, considering factors like safety, traffic rules, travel time, comfort, and customer satisfaction. Successful trips and high customer ratings indicate overall performance.

Environment (E): Dynamic and unpredictable urban road network, including road conditions, traffic flow, weather, pedestrians, vehicles, traffic signals, and obstacles. Real-time changes require effective navigation and adaptation.

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Actuators (A): Utilizes actuators like steering, acceleration, braking, turn signals, headlights, horn, and communication devices to interact with the environment. Performs tasks like lane changes, intersection stops, and passenger communication.

Sensors (S): Equipped with cameras, LiDAR, radar, GPS, ultrasonic sensors, IMUs, and microphones. Collects data on road conditions, traffic, positioning, pedestrians, weather, and more. Machine learning processes sensor data for informed driving decisions.

The PEAS description highlights key aspects of an autonomous taxis operation, emphasizing its performance objectives, dynamic urban environment, interaction mechanisms, and sensory perception capabilities.

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Agent's have varied environments (Russell & Norvig):

- Fully observable vs Partially observable
- Deterministic vs Stochastic
- Episodic vs Sequential
- Static vs Dynamic environments
- Discrete vs Continuous
- Single vs Multi-agent

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A1. Fully Observable Environment:

- In a fully observable environment, an agent's sensors provide unrestricted access to the entire state of the environment.
- The agent can perceive everything happening in its surroundings without ambiguity.
- No need for the agent to maintain internal memory or state to track the world; decisions rely solely on current sensory input.

Example: A robot in a room with cameras providing a complete view, enabling it to know the positions of all objects and obstacles.

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A2. Partially Observable Environment:

- A partially observable environment limits an agent's sensor access to the state of the environment.
- The agent cannot fully perceive or know the complete state, introducing uncertainty.
- Agents may need internal state or memory to track past observations and infer the current state.

Example: A vacuum-cleaning robot with a local dirt sensor that can't detect dirt beyond its vicinity, requiring exploration to identify dirt in other areas.

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Example: An automated taxi on the road unable to directly observe other drivers' thoughts or intentions, leading to partial observability.

Fully observable environments offer clarity, while partial observability requires handling uncertainty and maintaining internal state.



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B1. Deterministic Environment:

- In a deterministic environment, the outcome depends solely on the current state and the agent's action.
- There is no randomness; given the same initial conditions and actions, the environment always transitions to the same state.

Example: Chess played with a clock – each move is determined by player choices and game rules, resulting in a predictable outcome.

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B2. Stochastic Environment:

- In a stochastic environment, randomness or uncertainty influences the next state, even with known current state and action.
- The same action in the same situation may lead to different outcomes due to inherent randomness.

Example: Taxi driving – factors like traffic, weather, and other drivers introduce unpredictability, causing varying outcomes on the same route.

Thus, deterministic environments offer predictability, while stochastic environments require accounting for randomness in decision-making.

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C1. Episodic environments:

- Divides an agent's interaction into discrete episodes.
- Each episode involves receiving percepts and performing a single action.
- Outcomes in one episode don't affect those in future episodes.
- Common in tasks with no long-term consequences,

Example: a part-picking robot on a conveyor belt.

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C2. Sequential environments

- Dependencies occur between episodes.
- Current actions influence all future episodes.
- Actions in one episode can set the stage for future interactions.
- **Examples** include chess (each move affects the whole game) and taxi driving (route choices impact the entire journey).

Understanding episodic vs. sequential environments is vital for agent design and strategy, as the level of interdependence between episodes dictates decision complexity.

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D1. Static Environment:

- In a static environment, the environment's state remains constant during the agent's decision-making process.
- It is stable and unchanging, allowing the agent to assume the validity of its perceptions throughout decision-making.
- Agents don't need continuous monitoring or real-time updates.
- An **example** is a crossword puzzle, where clues and letters remain fixed.

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D2. Dynamic environment

- It undergoes changes while the agent makes decisions.
- It is unpredictable, requiring agents to adapt to evolving conditions.
- Agents must continuously monitor and adjust decisions in response to changes.
- For **instance**, taxi driving involves changing traffic conditions, obstacles, and unexpected events.

Understanding static vs. dynamic environments is vital for agent adaptation, as it affects the need for real-time updates and decision flexibility.

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E1. Discrete Environment

- In a discrete environment, states or actions are **finite and countable**.
- Represented by discrete variables or events with distinct values.

Example: A crossword puzzle where actions involve selecting specific letters or words from a finite set.

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E2. Continuous environment

- In a continuous environment, states or actions involve **uncountable** possibilities.
- Variables can take infinite values, and actions may be continuous movements or adjustments.

Example: Taxi driving with continuous coordinates, continuous control, and continuous time.

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F1. Single-agent Environment

- In a single-agent environment, there's only one agent interacting with the environment.
- The agent makes decisions independently.

Example: A crossword puzzle solver works alone.

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F2. Multi-agent Environment

- In a multi-agent environment, multiple agents interact.
- Agents' actions can affect and be affected by others.
- Can be competitive or cooperative.

Example: Chess (competitive), taxi driving (partially cooperative/competitive).

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- **Simple Reflex Agents:** Agents that make decisions based solely on the current percept without considering past history.

- **Model-Based Reflex Agents:** Agents that maintain an internal model of the world to make decisions based on both the current percept and past history.

- **Goal-Based Agents:** Agents that consider goals and objectives when making decisions, aiming to achieve specific outcomes.

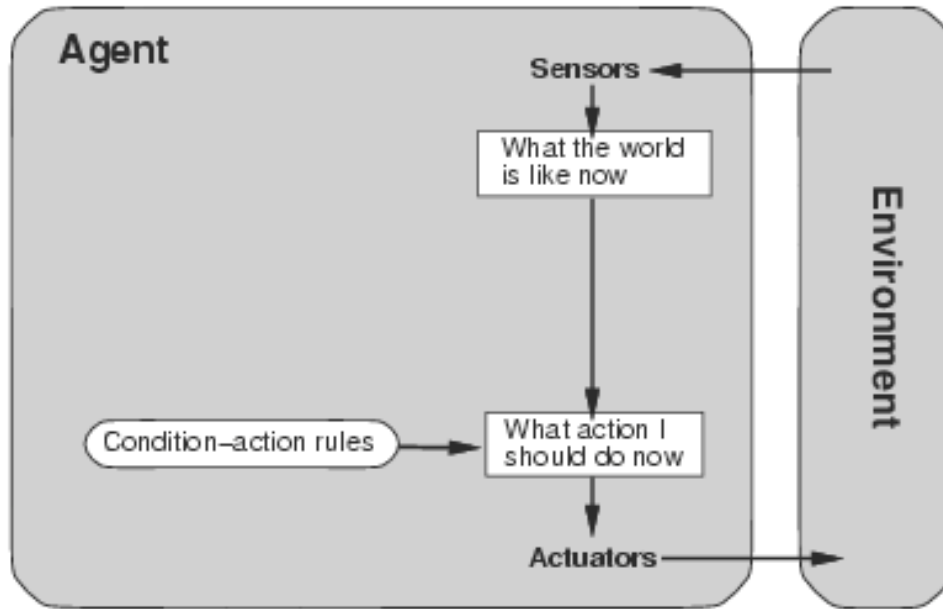
- **Utility-Based Agents:** Agents that make decisions by assigning utilities or values to different outcomes and selecting actions that maximize expected utility.

- **Learning Agents:** Agents that improve their decision-making abilities over time through learning from past experiences.

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Simple Reflex Agents (doesn't consider past history) eg: Thermostat



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- Basic type of artificial agent.
- Decide based only on current percept.
- No history/internal model considered.
- Actions follow simple if-then rules.

Examples: Thermostats (heating control).

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Key Characteristics:

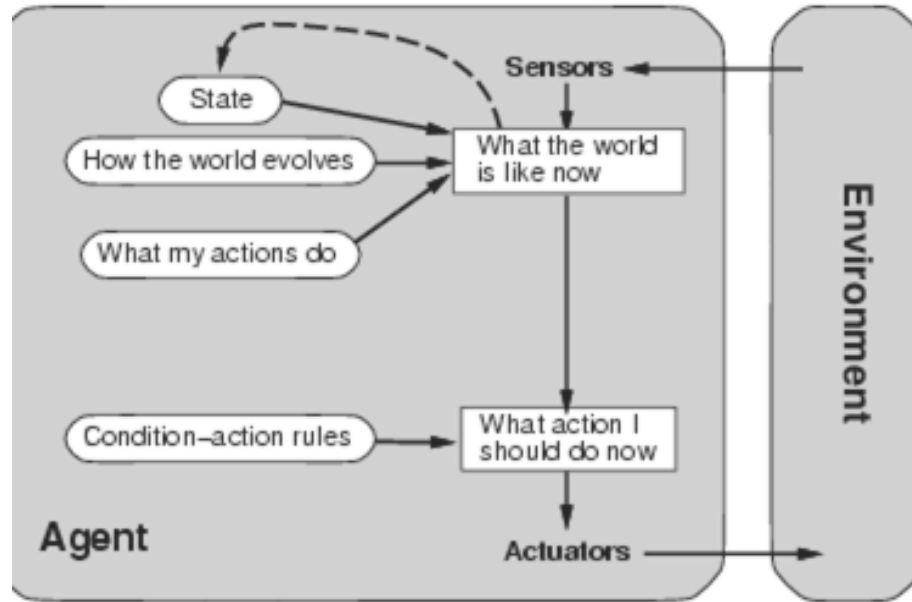
- **Immediate Response:** React instantly to current sensory input.
- **No Memory:** Lack memory of past percepts or actions.
- **Limited Scope:** Decisions follow fixed condition-action rules.
- **Lack of Adaptability:** No gradual learning or adaptation.

Simple Reflex Agents operate in the present, making decisions based on immediate sensory input and predefined rules.

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Model-Based Reflex Agents (current and past into account) - Eg: Autonomous vacuum cleaner



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- Maintain internal model of the environment.
- Consider both current percept and past history.
- Improve decision-making through historical context.
- Use condition-action rules, often with historical context.
- Some have limited learning capabilities.

Examples: Autonomous vacuum cleaner navigating room.

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Key Characteristics:

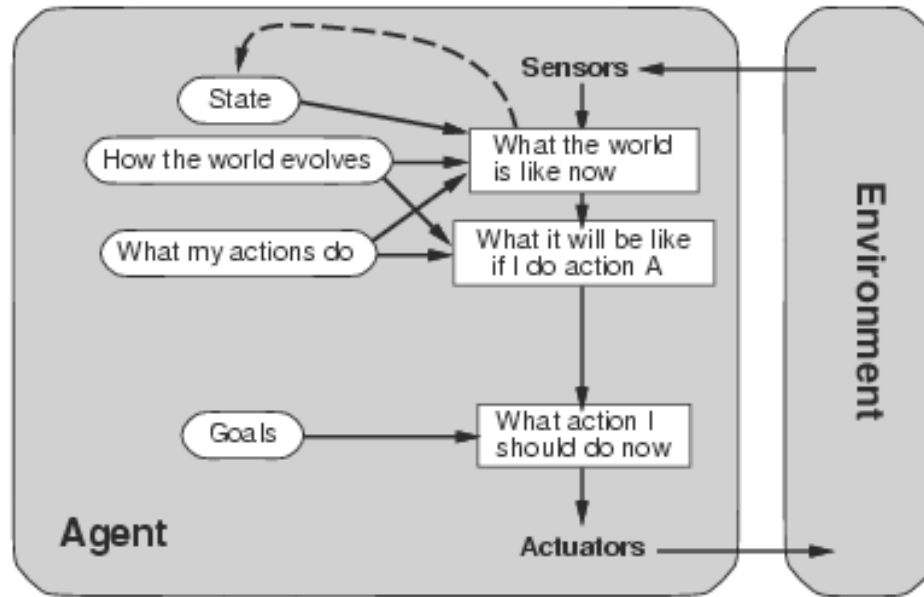
- **Internal Model:** Maintain an internal representation of the world.
- **History Consideration:** Take into account past sensory inputs and actions.
- **Improved Decision-Making:** Make context-aware and adaptive decisions.

Model-Based Reflex Agents enhance decision-making by maintaining an internal model that considers past history alongside current percepts.

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Goal-Based Agents (have a fixed goal/objective) - eg: Siri



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- Focus on achieving specific goals or objectives.
- Make decisions to reach desired outcomes.
- Use planning processes to determine action sequences.
- Adapt and change actions based on progress and situations.
- Consider long-term consequences and trade-offs.
- Some can learn from past experiences.

Examples: Personal assistant applications like Siri or Google Assistant.

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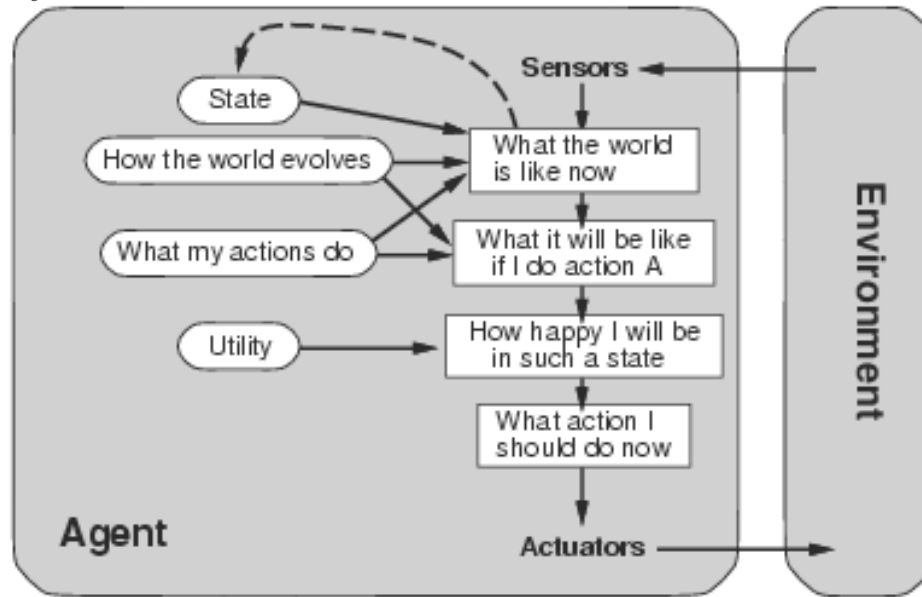
- **Goal-Oriented:** Driven by predefined goals or objectives.
- **Planning:** Engage in planning to achieve goals.
- **Flexibility:** Adapt to changing circumstances.

Goal-Based Agents aim to achieve specific objectives by planning and adapting actions, considering long-term consequences and trade-offs.

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Utility-Based Agents (assigns utilities of values) eg: Recommendation systems



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Utility-Based Agents

- Make decisions by maximizing expected utility.
- Assign utilities/values to outcomes.
- Use a **utility function to represent preferences**.
- Consider probabilities and handle risk.
- Adapt and make trade-offs based on preferences.
- Used in recommendation systems, finance, and more.

Examples: Recommendation systems, financial portfolio optimization.

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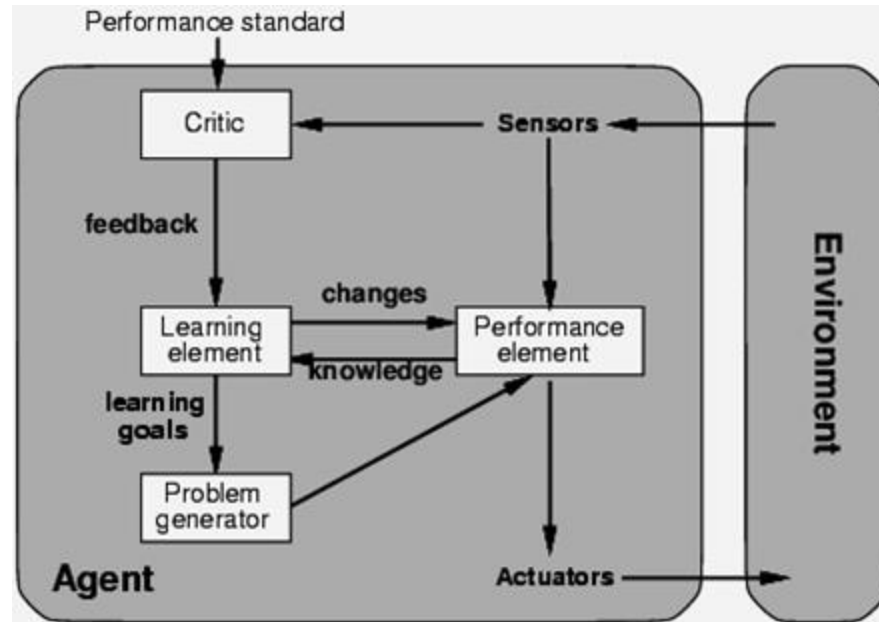
- **Utility Function:** Assigns numerical values to outcomes.
- **Decision-Making:** Maximizes expected utility for actions.
- **Risk Handling:** Considers probabilities and uncertainties.

Utility-Based Agents use utility functions to make decisions based on preferences, considering probabilities and adapting to changing circumstances.

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Learning Agents (improves their decision making) eg: Autonomous Vehicles



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Learning Agents

- Improve performance through learning.
- Adapt and adjust based on past experiences.
- Generalize knowledge to handle new situations.
- **Types:** Supervised, unsupervised, reinforcement learning.
- Used in various fields like NLP, robotics, recommendation systems.
- Continuous improvement and ethical challenges.

Examples: Autonomous vehicles.

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Key Characteristics:

- Learning from Data: Acquire knowledge from data or interactions.
- Adaptability: Adjust behavior based on accumulated experience.
- Generalization: Apply past learning to new, unseen situations.

Learning Agents continually improve their performance by learning from data and adapting to new situations, making them crucial in various applications but also presenting challenges in terms of ethical use and safety.

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- An agent is something that interacts with its environment by reading and taking actions.
- A performance measure assesses how well an agent is doing to achieve the desired outcome.
- When creating an agent, it's important to understand the environment it operates in.
- Agents and their programs are designed to perform specific tasks.
- Learning agents continuously adjust their components based on feedback to improve their overall performance.

Thank you